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Applied AI and Enablement Portfolio

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Selected projects from the past three years, organized around one working method: build the system, document what breaks, and turn the result into instruction. Each entry gives the problem, the approach, the stack, and current status. Full role history and credentials are in the accompanying resume.

AI ENABLEMENT, CURRICULUM, AND CERTIFICATES

AI-Epistemic Resilience (AI-ER): Framework, Book, and Certificate Course

Author and course designer | 2025 to present

Problem: As generative AI becomes a default reference tool, students and professionals need a structured way to check AI output against evidence before accepting it.

- **Framework:** Four interacting capacities: verification practices, epistemic humility, adaptive reasoning, and metacognitive monitoring, with epistemic regulation framed as a disciplined practice.
- **Book:** AI-Epistemic Resilience: A Framework for Knowledge Integrity in the Age of Artificial Intelligence (2026). A journal manuscript drawing on the framework is under review at Computers and Education Open.
- **Certificate course:** Converted the framework into a free, self-paced eight-module course with a final assessment and certificate, hosted at accesstorobotics.org and used by educators, healthcare professionals, and industry practitioners. The robot and co-instructor builds below serve as worked case studies, including a documented persona-authoring incident used to teach why AI output is an authored artifact that must be checked.

Status: Book and course published. Manuscript under peer review.

AI Fundamentals for Educators

Curriculum design for Franklin University | 2026

- Eight-week, cross-listed undergraduate and graduate course aligned to ISTE and InTASC standards, concluding with a capstone instructional unit each candidate can deploy in their own teaching context. Designed and delivered to the university; not adopted as of mid-2026.

Anthropic and Generative AI Credentials

2023 to 2026

- **Anthropic Academy course certificates (2026):** Claude 101, Claude Platform 101, Claude Code 101, AI Capabilities and Limitations.
- **Coursework (2023 to 2025):** Prompt engineering fundamentals, large language models, AI-first workflow design, and applied AI concepts; Franklin University cybersecurity training (2025). A continuous line from prompt engineering in 2023 to platform and agentic tooling in 2026.

AI CO-INSTRUCTORS IN LIVE INSTRUCTION

Voice-driven assistants built to support live teaching, each with its own persona architecture. All are independently built personal projects.

Aion Sage and GAIA: Academic Teaching Assistants

Persona-differentiated assistants | 2024 to present

- **Approach:** A persona architecture that separates the automation specialist from a multidisciplinary academic educator, each running on layered system instructions controlling tone, instructional style, and domain. Real-time avatar presence gives the assistant a visible classroom identity.
- **Classroom pilot:** Aion Sage was piloted as a co-lecturer in a university biology course, delivering content and answering students in real time alongside the instructor. Student response was strongly positive, with learners describing the interaction as engaging and immersive.

Stack: Python, Anthropic Claude and OpenAI APIs, VMagicMirror avatar rendering, Ollama and LM Studio for offline testing.

SIA: Instructor Assistant for Industrial Automation

Voice co-instructor for PLC labs | 2025 to present

- **Problem and approach:** In long automation labs the instructor is often the only person who can answer PLC questions in real time, which stalls practice for everyone else. SIA listens passively during live and virtual labs and responds only when explicitly invited, so it never competes with the instructor for the room.
- **Engineering:** Solved microphone-to-speaker feedback loops, designed half-duplex turn management so it does not talk over people, and tuned response timing to a classroom pace. Packaged as a standalone Windows build. A personal proof of concept piloted in my own classes; it is not a Siemens product.

Stack: Python, OpenAI Realtime WebSocket API, sounddevice streaming, PyInstaller.

Aurora and Bentley Chat

Realtime avatar assistant; custom Alexa skill | 2025 to present

- **Aurora:** An end-to-end realtime pipeline (wake-word activation, low-latency audio over WebSockets, avatar lip-sync through a virtual audio cable) with modes tuned for teaching, live demonstration, and technical support, packaged for classroom and lab machines and able to recover from network interruption.

- **Bentley Chat:** A custom Alexa skill with a defined persona and conversational memory, built to teach how conversational memory works. Both assistants are in active use.

Stack: Python, OpenAI Realtime API, Vosk wake-word detection, VB-CABLE, VMagicMirror, AWS, Alexa Skills Kit and APL.

LOCAL AND PRIVATE AI SYSTEMS

ReggieGPT: Local Retrieval-Augmented Grading Assistant

Local RAG deployment | 2026

- **Problem and approach:** Student work cannot be sent to cloud services. Deployed a private assistant on a Mac mini running a quantized 27B Qwen3 model through llama-server and Open WebUI, with course documents, rubrics, and reference cases loaded into a retrieval knowledge base so the model grounds its review in the actual grading criteria. No web access and no cloud fallback; files move between machines over the local network only.

Stack: llama.cpp / llama-server, Open WebUI knowledge base (chunking, local embeddings, vector store), Qwen3 27B 4-bit, macOS launchd.

Rose: Fully Local Companion Robot

Local inference and small-model persona engineering | 2026 to present

- **Local stack:** Speech recognition, a quantized 4B vision-language model, and speech synthesis all run on a Mac mini M4 with no cloud service in the loop, using roughly six of twenty-four gigabytes and synthesizing at a real-time factor of 0.26, fast enough for natural speech.
- **Small-model persona engineering:** A 4B-class model performs described behaviors out loud, so rules written as descriptions of silence produce a robot that narrates being quiet. Rewrote the instruction set around blunt behavioral directives, concrete examples, and explicitly labeled wrong answers. Structured memory lives outside the application so updates cannot destroy it; speaker recognition gates personal detail; services auto-respawn and the robot reconnects without human intervention.

Stack: Parakeet speech recognition, quantized Qwen3-VL 4B via llama.cpp, Kokoro synthesis with PSOLA voice transformation, Python, launchd, WebSocket transport.

EMBODIED AI: CLOUD COMPANION ROBOT

Aiden: Multi-Backend Cloud Companion

Systems architecture, memory design, reliability | 2026 to present

Aiden and Rose are two Reachy Mini robots built as a deliberate contrast, cloud reasoning versus fully local, so students can hear the deployment tradeoff in the room.

- **Backend architecture:** Interchangeable cloud realtime backends (a HuggingFace-hosted Qwen3 model, OpenAI gpt-realtime, Gemini Live) on a Raspberry Pi Compute Module 4, with a command-line utility that swaps provider, model, and endpoint, restarts the session through the daemon REST API, and verifies the reload from the system journal.
- **Memory architecture:** A three-stage pipeline separate from the context window: daily structured logs, LLM-driven extraction into a schema with date, tone, summary, and confidence, and a load stage merging extracted memories with a hand-curated permanent store. Diagnosed a recall defect that silently dropped curated entries and replaced it with a term-frequency relevance scorer.
- **Reliability and oversight:** Documented the systemd services, watchdogs, Wi-Fi failover, and a post-update patch manager; wrote recovery playbooks that separate reliable signals from persistent red herrings. The agent cannot edit its own persona, tools, or credentials and must obtain verbal acknowledgment before irreversible actions, applying the AI-ER human-oversight principles to an always-on physical system.

Stack: Python, systemd and inotify, ALSA and PipeWire with a LADSPA DSP stage, OpenAI Realtime API, HuggingFace realtime inference, Gemini Live API, JSONL logging, REST session management.

Status: In daily operation and used as a working case study in the AI-ER course.

OUTREACH AND INDUSTRY

The Finley Robotics Initiative

Founder, Georgia nonprofit | 2026 to present

- Robotics and AI-literacy program placing real hardware with students entering grades 9 through 11, paired with twelve months of monthly mentorship from practicing engineers and educators and a public project showcase. Delivers Smart Tools, Sharper Minds, a school and community series on large language models, responsible AI use, and disinformation resistance. The inaugural scholarship cycle is open, with applications closing November 30, 2026.

Sunday (sunday.ai): Embodied AI Data Contribution

Contract role | 2026 to present

- Capture real-world task demonstrations for teleoperation and embodied-AI training in home robotics.

References, source code samples, and live demonstrations are available on request.